SYNTHETIC PROCUREMENT RESPONSE | CLASSROOM DEMONSTRATION

# Apex Systems

Graduate Admissions Case Management & Document Workflow

Architecture-forward proposal | Higher cost | 20-week schedule

## 1. Executive summary

Apex proposes a single-tenant, managed admissions workspace with a relational case store, object storage for documents, and asynchronous integration workers. The design is intended to absorb the stated 1,200-user peak while separating applicant traffic from staff review queues. The offer prices all four named interfaces and three-cycle migration in the implementation fee.

## 2. Understanding of the university's requirements

Apex interprets the seven-year audit requirement as an operational and archival obligation: two years searchable in the application and five years exported nightly to the university's controlled archive. It assumes Morrowfield will provide Banner REST API credentials, Box enterprise app approval, a non-production Entra tenant, and representative migration extracts by week 3.

## 3. Proposed solution, architecture, and integrations

The service is proposed in a US East cloud region with separate production and non-production accounts. A queue-based adapter layer uses documented Banner REST resources, Entra SAML plus SCIM provisioning, Microsoft Graph for Exchange notices, and Box APIs for document references. Each adapter records an idempotency key, retry count, and reconciliation status; failed messages enter a review queue rather than being silently discarded. The proposal includes 12 named data mappings and a daily integration reconciliation report.

**COMMON REQUEST - IDENTICAL FOR ALL FOUR SUPPLIERS**

Morrowfield University is a fictional public university seeking a secure cloud service for graduate admissions case management. The same scope applies to every response: 18,000 applications per year across 11 academic divisions; 55 staff users; peak loads near 1,200 concurrent portal users; migration of three prior cycles (up to 54,000 applicant records and approximately 400,000 documents); and a production launch target of 30 April 2027 after a 26-week delivery window beginning 2 November 2026. Required interfaces are Ellucian Banner student information, Microsoft Entra ID, Exchange Online, and Box. Deliverables include an applicant portal, configurable review queues, document checklist, decision notifications, role-based staff workspace, reports, migration, training, and cutover support. The security baseline asks for TLS 1.2 or stronger, AES-256 encryption at rest, MFA/SSO, least-privilege access, auditable decisions, a seven-year audit-retention plan, US data residency, annual independent penetration testing, and incident notice within 24 hours. The university requests a three-year total cost of ownership, stated assumptions, and a support plan.

## 4. Implementation plan, timeline, staffing, and responsibilities

The schedule below is the supplier's proposed sequence. Dates remain dependent on timely university decisions, access to test tenants, and acceptance of the stated scope.

| Window | Milestone | Activities and exit evidence |
|---|---|---|
| Weeks 1-2 | Discovery and tenant readiness | Confirm data owners, API access, security boundary, and acceptance criteria. |
| Weeks 3-6 | Architecture and configuration | Approve field maps, roles, retention export, and portal prototype. |
| Weeks 7-12 | Build and interfaces | Configure workflow; implement and test four adapters in a shared sandbox. |
| Weeks 13-16 | Migration and UAT | Reconcile three cycles; run load, accessibility, and admissions-team testing. |
| Weeks 17-20 | Cutover and hypercare | Train 55 staff; launch in waves; provide 10 business days of hypercare. |

## Proposed delivery team

| Role | FTE | Responsibility |
|---|---|---|
| Engagement lead | 1.0 | Delivery governance, dependencies, and weekly decision log. |
| Solution architect | 1.0 | Security design, data model, and technical approvals. |
| Integration engineers | 2.0 | Four adapters, retry handling, and reconciliation. |
| Configuration / QA | 1.5 | Workflow configuration, regression, and load testing. |
| Security / change | 0.7 | Control evidence, role review, training, and adoption. |

## Integration and delivery dependencies

The service is proposed in a US East cloud region with separate production and non-production accounts. A queue-based adapter layer uses documented Banner REST resources, Entra SAML plus SCIM provisioning, Microsoft Graph for Exchange notices, and Box APIs for document references. Each adapter records an idempotency key, retry count, and reconciliation status; failed messages enter a review queue rather than being silently discarded. The proposal includes 12 named data mappings and a daily integration reconciliation report.

## 5. Commercial proposal

| Cost element | Amount (USD) | Basis |
|---|---|---|
| Implementation, migration, and four interfaces | $186,000 | One-time |
| Managed subscription and standard support | $114,000 | Per year |
| Three-year total cost of ownership | $528,000 | $186,000 + 3 x $114,000 |

Pricing includes the four interfaces as described, migration of up to 54,000 records and 400,000 documents, two training cohorts, and standard hosting. Taxes, a fifth interface, custom document remediation, and more than 10 days of post-launch hypercare are excluded. Subscription assumes annual prepayment and no more than 20% annual growth in stored documents.

## 6. Security, compliance, privacy, auditability, and risk

The synthetic control package states TLS 1.2+ in transit, AES-256 encryption at rest with managed key rotation, Entra MFA/SCIM, quarterly privileged-access review, and immutable administrator and decision logs. The proposal describes a fictional SOC 2 Type II report covering the hosting service and a penetration test completed in July 2026; both are demonstration claims and would require independent verification. The platform keeps two years of searchable audit events and exports the remaining five years nightly to the university archive. Stated recovery objectives are RPO 1 hour and RTO 4 hours. A material risk is the university's timely provision of archive storage and identity approvals.

## 7. Support arrangements, experience, references, and limitations

Standard support is staffed 08:00-18:00 Eastern on business days, with a 30-minute acknowledgement target for P1 incidents and a four-hour workaround target. After-hours P1 coverage is on-call, not a continuously staffed desk. Apex cites the fictional Westbridge College rollout (11,200 annual applications, 9 integrations) and a fictional South Quay Institute pilot (7,500 applications). References use reserved `.invalid` contact domains and are synthetic.

## Risks and limitations

| The 20-week plan leaves six weeks of schedule reserve, but API approvals are on the critical path.

| Seven-year retention depends on a customer-managed archive export being monitored by Morrowfield.

All supplier organizations, employees, customers, references, contact details, attestations, prices, and performance claims in this PDF are fictional synthetic demonstration data. They are not offers, independently verified evidence, or real supplier records. Reference domains intentionally use the reserved `.invalid` suffix.